Lecture 24

Lorentz groups and finite-dimensional representations

Objectives. *We build the double cover* $\mathrm{SL}(2,\mathbb{C}) \to \mathrm{SO}^+(1,3)$ *from Hermitian* 2×2 *matrices. We then classify the finite-dimensional irreducible continuous complex representations of both groups by complexifying the real Lie algebra.*

By the end of this lecture you will be able to:

- Prove that $\mathrm{SL}(2,\mathbb{C}) \to \mathrm{SO}^+(1,3)$ is a surjective covering homomorphism with kernel $\{\pm I\}$, and identify $\mathrm{SL}(2,\mathbb{R})$ as the double cover of $\mathrm{SO}^+(1,2)$ inside it.
- Identify $\mathfrak{sl}(2,\mathbb{C})_{\mathbb{R}} \otimes_{\mathbb{R}} \mathbb{C}$ with $\mathfrak{sl}(2,\mathbb{C}) \oplus \mathfrak{sl}(2,\mathbb{C})$.
- Label the irreducible finite-dimensional continuous complex representations of $\mathrm{SL}(2,\mathbb{C})$ by (j_L, j_R) and decide when they descend to $\mathrm{SO}^+(1,3)$.
- Recognise Weyl spinors and four-vectors in this list.

In special relativity an event has coordinates $x = (x_0, \vec{x})$, with time x_0 and position $\vec{x} \in \mathbb{R}^3$. We use units in which light has speed 1. Inertial observers are related by linear maps that preserve the Minkowski form below. Relativistic fields take values in finite-dimensional representations of the Lorentz group or its double cover $\mathrm{SL}(2,\mathbb{C})$.

24.1 Minkowski space and the Lorentz group

Let $e_0, \dots, e_3$ be the standard basis of $\mathbb{R}^4$. The *Minkowski form* is

$$\langle x, y\rangle = x_0 y_0 - \vec{x}\cdot\vec{y} = x^{T}\eta y, \qquad \eta = \mathrm{diag}(1,-1,-1,-1).$$

Definition 24.1. The *Lorentz group* is $\mathrm{O}(1,3) = \{\Lambda \in \mathrm{GL}(4,\mathbb{R}) : \Lambda^{T}\eta\Lambda = \eta\}$. The *future cone* is $T^+ = \{x \in \mathbb{R}^4 : x_0 > |\vec{x}|\}$.

Let $\Lambda \in \mathrm{O}(1,3)$. Then $\det\Lambda = \pm 1$ and $\Lambda^{-1} = \eta\Lambda^{T}\eta$, so $(\Lambda^{-1})_{00} = \Lambda_{00}$. Since $\langle \Lambda e_0, \Lambda e_0\rangle = 1$,

$$\Lambda_{00}^2 = 1 + \Lambda_{10}^2 + \Lambda_{20}^2 + \Lambda_{30}^2 \geq 1.$$

So either $\Lambda_{00} \geq 1$ or $\Lambda_{00} \leq -1$.

Lemma 24.2. *Let* $\Lambda \in \mathrm{O}(1,3)$. *Then* $\Lambda_{00} \geq 1$ *if and only if* $\Lambda T^+ \subseteq T^+$. *Consequently the elements with* $\Lambda_{00} \geq 1$ *form a subgroup* $\mathrm{O}^+(1,3)$.

Proof. Suppose $\Lambda_{00} \geq 1$ and let $x \in T^+$. Put $w = \Lambda^{-1} e_0$. Then $\langle w, w\rangle = 1$ and $w_0 = (\Lambda^{-1})_{00} = \Lambda_{00} \geq 1$, so $w_0 > |\vec{w}|$. Since Λ preserves the form,

$$(\Lambda x)_0 = \langle e_0, \Lambda x\rangle = \langle w, x\rangle = w_0 x_0 - \vec{w}\cdot\vec{x} \geq w_0 x_0 - |\vec{w}|\,|\vec{x}| > 0.$$

The last step uses $w_0 > |\vec{w}|$ and $x_0 > |\vec{x}|$. Also $\langle \Lambda x, \Lambda x\rangle = \langle x, x\rangle > 0$. These two facts give $\Lambda x \in T^+$. Conversely, if $\Lambda T^+ \subseteq T^+$, then $\Lambda e_0 \in T^+$. So $\Lambda_{00} > 0$, hence $\Lambda_{00} \geq 1$. The cone condition is preserved by products, and $(\Lambda^{-1})_{00} = \Lambda_{00}$. So these elements form a subgroup. ∎

Definition 24.3. The *proper orthochronous Lorentz group* is

$$\mathrm{SO}^+(1,3) = \{\Lambda \in \mathrm{O}(1,3) : \det\Lambda = 1,\ \Lambda_{00} \geq 1\}.$$

By the cone lemma, Lemma 24.2, it is a group. It is closed in $\mathrm{GL}(4,\mathbb{R})$, hence a Lie group by the closed-subgroup theorem (Lecture 3). By the two dichotomies above, for $\det\Lambda$ and for Λ_{00}, it is also open in $\mathrm{O}(1,3)$. So it has the Lie algebra of $\mathrm{O}(1,3)$:

$$\mathfrak{so}(1,3) = \{Y \in \mathfrak{gl}(4,\mathbb{R}) : Y^T\eta + \eta Y = 0\}.$$

The map $Y \mapsto \eta Y$ is a bijection onto the antisymmetric matrices, so $\dim\mathfrak{so}(1,3) = 6$.

A basis consists of the *rotation generators* J_k and the *boost generators* K_k, for $k = 1,2,3$. Here $J_k e_0 = 0$, and J_k acts on $\vec{x}$ by $e_k \times \vec{x}$. Moreover $K_k e_0 = e_k$ and $K_k e_k = e_0$, and K_k maps the other two basis vectors to zero. For example,

$$J_3 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad K_3 = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}.$$

Then $R_3(\theta) = \exp(\theta J_3)$ rotates the (x_1, x_2)-plane by the positive angle θ. The *boost* $B_3(t) = \exp(tK_3)$ fixes x_1, x_2 and acts by

$$x_0 \pm x_3 \mapsto e^{\pm t}(x_0 \pm x_3).$$

The quantity $x_0^2 - x_3^2 = (x_0 + x_3)(x_0 - x_3)$ does not change, so a point moves along a hyperbola. Figure 24.1 shows this motion in the plane $x_1 = x_2 = 0$.

Parity $\mathrm{diag}(1,-1,-1,-1)$ and time reversal $\mathrm{diag}(-1,1,1,1)$ lie in $\mathrm{O}(1,3)$ but outside the open subgroup $\mathrm{SO}^+(1,3)$. The Lie algebra $\mathfrak{so}(1,3)$ carries no information about these elements. We study only $\mathrm{SO}^+(1,3)$.

24.2 The double cover

Lecture 4 built the covering $\mathrm{SU}(2) \to \mathrm{SO}(3)$ as the adjoint action on $\mathfrak{su}(2)$. It also identified this with the action $\xi \mapsto U\xi U^*$ on traceless Hermitian matrices. We now allow all Hermitian matrices and all of $\mathrm{SL}(2,\mathbb{C})$, with the action $X \mapsto AXA^*$. The same construction gives the Lorentz group.

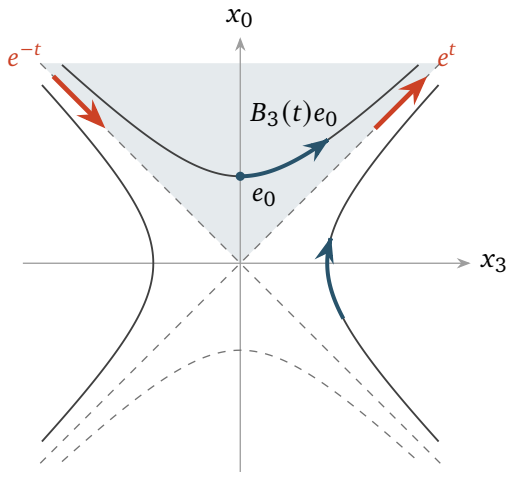


Figure 24.1: The boost $B_3(t)$ in the plane $x_1 = x_2 = 0$. The dashed diagonals are the null lines $x_0 = \pm x_3$, and the shaded wedge is the slice of T^+. The solid curves are the hyperbolas $x_0^2 - x_3^2 = \pm 1$. The coloured arrows show that $B_3(t)$ stretches the diagonal $x_0 = x_3$ by e^t and shrinks the diagonal $x_0 = -x_3$ by e^{-t}. The heavy arcs show the motion of two points for one fixed $t > 0$: the point e_0, which is the lowest point of the upper branch, and a point with $x_0^2 - x_3^2 = -1$. The dashed lower branch is the slice of the sheet on which Λe_0 lies when $\Lambda_{00} \le -1$.

Let $\sigma_0 = I$ and

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

These form a real basis of the space $\mathcal{H}$ of Hermitian 2×2 matrices. For $x \in \mathbb{R}^4$ put

$$X(x) = \sum_{\mu=0}^{3} x_\mu \sigma_\mu = \begin{pmatrix} x_0 + x_3 & x_1 - i x_2 \\ x_1 + i x_2 & x_0 - x_3 \end{pmatrix}.$$

Direct expansion gives

$$\det X(x) = \langle x, x \rangle, \qquad x_0 = \tfrac{1}{2} \operatorname{tr} X(x).$$

For $A \in \mathrm{SL}(2, \mathbb{C})$ the matrix $AX(x)A^*$ is again Hermitian. Define $\varphi(A) \in \mathrm{GL}(4, \mathbb{R})$ by

$$X(\varphi(A)x) = A\, X(x)\, A^*. \tag{24.1}$$

The entries of $\varphi(A)$ are real polynomials in the real and imaginary parts of the entries of A. Hence φ is smooth.

Example 24.4 (Diagonal elements). Let $A = \operatorname{diag}(a, a^{-1})$ with $a \in \mathbb{C}^\times$. Then

$$AX(x)A^* = \begin{pmatrix} |a|^2 (x_0 + x_3) & (a/\bar{a})(x_1 - i x_2) \\ (\bar{a}/a)(x_1 + i x_2) & |a|^{-2}(x_0 - x_3) \end{pmatrix}.$$

For $a = e^{-i\theta/2}$, the coordinates x_0, x_3 are fixed and $x_1 + i x_2 \mapsto e^{i\theta}(x_1 + i x_2)$. For $a = e^{t/2}$, we get $x_0 \pm x_3 \mapsto e^{\pm t}(x_0 \pm x_3)$, with x_1, x_2 fixed. Put $X_k = -\frac{i}{2}\sigma_k$. Then

$$\varphi(\exp \theta X_3) = R_3(\theta), \qquad \varphi(\exp t i X_3) = B_3(t).$$

Differentiating at 0 gives $d\varphi(X_3) = J_3$ and $d\varphi(iX_3) = K_3$. Also $\operatorname{diag}(e^{i\theta}, e^{-i\theta}) = \exp(-2\theta X_3)$ maps to $R_3(-2\theta)$, as in the convention for $\mathrm{SU}(2) \to \mathrm{SO}(3)$ in Lecture 4. Finally, $\exp(2\pi X_3) = -I$ maps to $R_3(2\pi) = I$.

Let $\mathfrak{sl}(2,\mathbb{C})_{\mathbb{R}}$ denote $\mathfrak{sl}(2,\mathbb{C})$ regarded as a six-dimensional real Lie algebra.

Theorem 24.5. *The map φ is a surjective homomorphism $\mathrm{SL}(2,\mathbb{C}) \to \mathrm{SO}^+(1,3)$ with kernel $\{\pm I\}$. Its differential $d\varphi : \mathfrak{sl}(2,\mathbb{C})_{\mathbb{R}} \to \mathfrak{so}(1,3)$ is an isomorphism of real Lie algebras, and φ is a two-sheeted covering map.*

Proof. Step 1: a homomorphism into $\mathrm{O}^+(1,3)$. Since $(AB)X(AB)^* = A(BXB^*)A^*$, we have $\varphi(AB) = \varphi(A)\varphi(B)$. Since $\det A = 1$, $\det(AXA^*) = \det X$. So $\varphi(A)$ preserves $\langle x, x\rangle$. By polarisation it preserves $\langle x, y\rangle$, so $\varphi(A) \in \mathrm{O}(1,3)$. Its 00 entry is the time component of AIA^*:

$$\varphi(A)_{00} = \tfrac{1}{2}\operatorname{tr}(AA^*) = \tfrac{1}{2}\textstyle\sum_{i,j} |A_{ij}|^2 > 0.$$

By the dichotomy above, $\varphi(A)_{00} \geq 1$.

Step 2: determinant one. We show that $\mathrm{SL}(2,\mathbb{C})$ is path connected. Let $A \in \mathrm{SL}(2,\mathbb{C})$. Let $P = (A^*A)^{1/2}$ be the positive definite square root, and $U = AP^{-1}$. Then $U^*U = I$ and $\det P = |\det A| = 1$, so $U \in \mathrm{SU}(2)$. Diagonalise U and P by the spectral theorem. Since both have determinant 1, we can write $U = \exp Y_1$ and $P = \exp Y_2$ with Y_1 skew-Hermitian, Y_2 Hermitian, and both traceless. Then $t \mapsto \exp(tY_1)\exp(tY_2)$, for $t \in [0,1]$, is a path in $\mathrm{SL}(2,\mathbb{C})$ from I to A. Along it, $\det\varphi$ is continuous with values in $\{\pm 1\}$, and it starts at 1. So $\det\varphi(A) = 1$.

Step 3: the kernel. Suppose $\varphi(A) = I$. Then $AXA^* = X$ for all $X \in \mathcal{H}$. Taking $X = I$ gives $AA^* = I$, so $AXA^{-1} = X$ for all $X \in \mathcal{H}$. Every complex matrix is $X_1 + iX_2$ with $X_1, X_2 \in \mathcal{H}$. So A commutes with all of $M_2(\mathbb{C})$, hence $A = cI$. Then $\det A = 1$ gives $c = \pm 1$. Conversely, $\pm I$ lie in the kernel.

Step 4: surjectivity. This is the main step. Let $\Lambda \in \mathrm{SO}^+(1,3)$ and $v = \Lambda e_0$. First we remove the boost part. The matrix $V = X(v)$ satisfies

$$\det V = \langle v, v\rangle = 1, \qquad \operatorname{tr} V = 2\Lambda_{00} > 0.$$

Its real eigenvalues have product 1 and positive sum, so both are positive. Thus $V = W\operatorname{diag}(\lambda, \lambda^{-1})W^*$ with W unitary and $\lambda > 0$. Put $P = W\operatorname{diag}(\lambda^{1/2}, \lambda^{-1/2})W^*$. Then $P \in \mathrm{SL}(2,\mathbb{C})$, $P^* = P$ and $P^2 = V$. Hence $X(\varphi(P)e_0) = PIP^* = V$, that is, $\varphi(P)e_0 = v$.

Next we treat the rotation part. Put

$$\Lambda' = \varphi(P)^{-1}\Lambda.$$

This element lies in $\mathrm{SO}^+(1,3)$ and fixes e_0. So it preserves $e_0^\perp = \operatorname{span}(e_1, e_2, e_3)$, where $-\langle\cdot,\cdot\rangle$ is the Euclidean inner product. Hence $\Lambda' = \operatorname{diag}(1, R)$ with $R \in \mathrm{SO}(3)$. Let $n = Re_3$, a unit vector, and set $N = \sum_k n_k\sigma_k$. This Hermitian matrix satisfies

$$\operatorname{tr} N = 0, \qquad \det N = -|n|^2 = -1,$$

so the eigenvalues of N are ± 1. By the spectral theorem, $N = W'\sigma_3 W'^*$ with W' unitary. We correct the determinant of W' by replacing W' with

$$W'\operatorname{diag}(1, \overline{\det W'}).$$

Diagonal matrices commute with σ_3, so this replacement preserves the diagonalisation of N. The corrected W' belongs to $\mathrm{SU}(2)$.

Then $\varphi(W')e_0 = e_0$, since $W'W'^* = I$. Also $\varphi(W')e_3 = n$, since $W'\sigma_3 W'^* = N$. So $\varphi(W')^{-1}\Lambda'$ fixes e_0 and e_3. It acts on $\operatorname{span}(e_1, e_2)$ by an element of $\mathrm{SO}(2)$, so it equals $R_3(\theta)$ for some θ. By the rotation formula of Example 24.4,

$$\Lambda = \varphi(P)\,\varphi(W')\,R_3(\theta) = \varphi\big(PW' \exp(\theta X_3)\big).$$

In particular, φ maps $\mathrm{SU}(2)$ onto the rotations $\operatorname{diag}(1, R)$.

Step 5: the differential and the covering. Differentiate (24.1) along $\exp(tY)$, using $\exp(tY)^* = \exp(tY^*)$:

$$X(d\varphi(Y)x) = YX(x) + X(x)Y^*, \qquad Y \in \mathfrak{sl}(2,\mathbb{C}). \tag{24.2}$$

Suppose $d\varphi(Y) = 0$. Taking $X = I$ gives $Y^* = -Y$. Then $YX - XY = 0$ for all $X \in \mathcal{H}$, so Y is scalar, as in Step 3. Since $\operatorname{tr} Y = 0$, we get $Y = 0$. Thus $d\varphi$ is injective. Both algebras have real dimension 6, so $d\varphi$ is a linear bijection. The differential of a Lie group homomorphism preserves brackets, so $d\varphi$ is an isomorphism of real Lie algebras.

By Steps 2 and 4, $\mathrm{SO}^+(1,3) = \varphi(\mathrm{SL}(2,\mathbb{C}))$ is connected. A continuous surjective homomorphism of connected Lie groups with bijective differential is a covering map (Lecture 4). Its fibres are cosets of $\ker\varphi = \{\pm I\}$, so it has two sheets. ∎

Formula (24.2) gives $d\varphi(X_k) = J_k$ and $d\varphi(iX_k) = K_k$ for every k. The case $k = 3$ is the diagonal computation of Example 24.4, Exercise 24.1 treats $k = 1$, and the case $k = 2$ is identical. The brackets $[X_i, X_j] = \epsilon_{ijk}X_k$ then transfer through $d\varphi$:

$$[J_i, J_j] = \epsilon_{ijk}J_k, \qquad [J_i, K_j] = \epsilon_{ijk}K_k, \qquad [K_i, K_j] = -\epsilon_{ijk}J_k.$$

Steps 2 and 4 of the proof are the Cartan decomposition of Lecture 20, for the real algebra $\mathfrak{sl}(2,\mathbb{C})_{\mathbb{R}}$ with $\mathfrak{k} = \mathfrak{su}(2)$ and $\mathfrak{p} = i\,\mathfrak{su}(2)$. Step 2 writes $A = UP$, with $U \in \mathrm{SU}(2)$ and $P = \exp Y_2$ for a traceless Hermitian Y_2. So $\varphi(A) = \varphi(U)\varphi(P)$, which is a boost followed by a rotation. Step 4 writes the same kind of element in the other order, as a rotation followed by a boost.

The same construction one dimension down

The construction also works for the real subgroup $\mathrm{SL}(2,\mathbb{R})$ of $\mathrm{SL}(2,\mathbb{C})$. It then relates $\mathrm{SL}(2,\mathbb{R})$ to a Lorentz group with two space coordinates. Let $\mathbb{R}^{1,2}$ be the subspace $\{x \in \mathbb{R}^4 : x_2 = 0\}$, with the form $x_0^2 - x_1^2 - x_3^2$. Let $\mathrm{SO}^+(1,2)$ be the group of linear maps of $\mathbb{R}^{1,2}$ that preserve this form, have determinant one and satisfy $\Lambda_{00} \geq 1$.

> **Proposition 24.6.** *Each $\varphi(A)$ with $A \in \mathrm{SL}(2,\mathbb{R})$ fixes e_2 and preserves $\mathbb{R}^{1,2}$. Restriction to $\mathbb{R}^{1,2}$ gives a surjective homomorphism $\mathrm{SL}(2,\mathbb{R}) \to \mathrm{SO}^+(1,2)$ with kernel $\{\pm I\}$.*

Proof. Let A be real, so $A^* = A^T$. If $x_2 = 0$, then $X(x)$ is real symmetric, and so is $AX(x)A^T$. Hence $\varphi(A)$ preserves $\mathbb{R}^{1,2}$. Next, $X(e_2) = \sigma_2$, and $A\sigma_2 A^T = (\det A)\sigma_2 = \sigma_2$, so $\varphi(A)$ fixes e_2. Thus $\varphi(A)$ is the identity on $\mathbb{R}e_2$ and its restriction on $\mathbb{R}^{1,2}$. By Theorem 24.5, this restriction preserves the form, has determinant one and has $\Lambda_{00} \geq 1$. If the restriction is the identity, then $\varphi(A) = I$, so $A = \pm I$.

For surjectivity we repeat Step 4 with real matrices. Let $\Lambda \in \mathrm{SO}^+(1,2)$ and $v = \Lambda e_0$. The matrix $V = X(v)$ is real symmetric with positive eigenvalues and determinant one. A real orthogonal diagonalisation gives a real positive definite P with $\det P = 1$ and $P^2 = V$, so $\varphi(P)e_0 = v$. Then $\varphi(P)^{-1}\Lambda$ fixes e_0, so it acts on the (x_3, x_1)-plane by an element of $\mathrm{SO}(2)$. Direct multiplication shows that the rotation matrix with first column $(\cos\theta, \sin\theta)^T$ fixes e_0 and rotates the (x_3, x_1)-plane through the angle 2θ. So this element is $\varphi(k)$ for a rotation matrix k, and $\Lambda = \varphi(Pk)$ on $\mathbb{R}^{1,2}$. ∎

So Lecture 21, Lecture 22 and Lecture 23 studied the double cover of the Lorentz group of two space coordinates and one time. This is why Bargmann's paper, cited in Lecture 23, has the title *Irreducible unitary representations of the Lorentz group*.

The earlier models of that block appear here as well. The real matrices $\frac{1}{2}\sigma_3$ and $\frac{1}{2}\sigma_1$ map to the boost generators K_3 and K_1. The stabiliser of e_0 in $\mathrm{SL}(2,\mathbb{R})$ is $\{A : AA^T = I\}$, which is the subgroup K of rotation matrices. The proof above shows that the group acts transitively on the upper sheet $x_0^2 - x_1^2 - x_3^2 = 1$ with $x_0 > 0$. So this sheet is $\mathrm{SL}(2,\mathbb{R})/K$. As a homogeneous space it is the half-plane of Lecture 20 and the disk of Lecture 23. Finally, a real line $\mathbb{R}v$ in $\mathbb{R}^2$ goes to the matrix vv^T. This matrix has determinant zero and trace $|v|^2 > 0$, so it is a future light-like vector, and $A(vv^T)A^T = (Av)(Av)^T$. Thus the circle of real lines in Lecture 22 is the circle of light rays in $\mathbb{R}^{1,2}$.

24.3 Complexifying the Lorentz algebra

This section rests on one change of basis. In the complexified Lorentz algebra $\mathfrak{so}(1,3) \otimes_{\mathbb{R}} \mathbb{C}$, the elements $\frac{1}{2}(J_k - iK_k)$ obey the brackets of $\mathfrak{su}(2)$. So do the elements $\frac{1}{2}(J_k + iK_k)$, and the two sets commute. We now derive this.

By the double-cover theorem, Theorem 24.5, $\mathfrak{so}(1,3) \cong \mathfrak{sl}(2,\mathbb{C})_{\mathbb{R}}$. A complex representation of this real algebra need not be complex linear. For example, $Y \mapsto \bar{Y}$ is a real Lie algebra automorphism that is not complex linear. Complexification treats both kinds together.

Let $\mathfrak{g}$ be a real Lie algebra and V a finite-dimensional complex space. A real Lie algebra map $\rho : \mathfrak{g} \to \mathfrak{gl}(V)$ extends uniquely to a complex-linear Lie algebra map $\rho_{\mathbb{C}}(Y \otimes z) = z\rho(Y)$ on $\mathfrak{g} \otimes_{\mathbb{R}} \mathbb{C}$. A complex subspace is ρ-invariant if and only if it is $\rho_{\mathbb{C}}$-invariant. A linear map intertwines two such maps ρ if and only if it intertwines their extensions.

The real algebra $\mathfrak{sl}(2,\mathbb{C})_{\mathbb{R}}$ carries its own operator M, given by $M(Y) = iY$. It satisfies $M^2 = -1$ and $[MY, Z] = M[Y, Z]$. After complexifying, $M \otimes 1$ has the eigenvalues i and $-i$. Its eigenspaces are ideals, since $[(M \otimes 1)u, v] = (M \otimes 1)[u, v]$. They commute: if u and v have eigenvalues i and $-i$, then $i[u, v] = (M \otimes 1)[u, v] = -i[u, v]$. The next proposition identifies these ideals with two copies of $\mathfrak{sl}(2,\mathbb{C})$. Under it, $M \otimes 1$ becomes multiplication by i on the first copy and by $-i$ on the second.

Proposition 24.7. *The complex-linear map*

$$\iota : \mathfrak{sl}(2,\mathbb{C})_{\mathbb{R}} \otimes_{\mathbb{R}} \mathbb{C} \to \mathfrak{sl}(2,\mathbb{C}) \oplus \mathfrak{sl}(2,\mathbb{C}), \qquad Y \otimes z \mapsto (zY,\ z\bar{Y}),$$

is an isomorphism of complex Lie algebras.

Proof. The pair $(zY, z\bar{Y})$ is real bilinear in (Y, z), so ι is well defined and complex linear. Complex conjugation of matrices preserves brackets, so

$$\iota([Y \otimes z, Y' \otimes z']) = \big(zz'[Y,Y'],\ zz'[\bar{Y},\bar{Y}']\big) = [\iota(Y \otimes z), \iota(Y' \otimes z')].$$

Both Lie algebras have complex dimension 6, so it suffices to prove that ι is injective. A general element $Y \otimes 1 + Y' \otimes i$ has image $(Y + iY',\ \bar{Y} + i\bar{Y}')$. If this is zero, then $Y = -iY'$ and $\bar{Y} = -i\bar{Y}'$. Conjugating the first equation gives $\bar{Y} = i\bar{Y}'$. Hence $\bar{Y}' = 0$, so $Y' = 0$ and $Y = 0$. ∎

Since $\overline{iX_k} = -i\bar{X}_k$, we have $\iota(X_k \otimes 1) = (X_k, \bar{X}_k)$ and $\iota(iX_k \otimes i) = (-X_k, \bar{X}_k)$. Hence the elements

$$X_k^L = \tfrac{1}{2}(X_k \otimes 1 - iX_k \otimes i), \qquad X_k^R = \tfrac{1}{2}(X_k \otimes 1 + iX_k \otimes i)$$

satisfy $\iota(X_k^L) = (X_k, 0)$ and $\iota(X_k^R) = (0, \bar{X}_k)$. In other words, X_k^L and X_k^R are the images of $X_k \otimes 1$ under the projections $\frac{1}{2}(1 - iM \otimes 1)$ and $\frac{1}{2}(1 + iM \otimes 1)$ onto the two ideals. We identify $\mathfrak{sl}(2,\mathbb{C})_{\mathbb{R}} \otimes_{\mathbb{R}} \mathbb{C}$ with $\mathfrak{so}(1,3)_{\mathbb{C}} = \mathfrak{so}(1,3) \otimes_{\mathbb{R}} \mathbb{C}$ through the complex-linear extension of $d\varphi$. Then $X_k^L = \frac{1}{2}(J_k - iK_k)$ and $X_k^R = \frac{1}{2}(J_k + iK_k)$. These elements satisfy

$$[X_i^L, X_j^L] = \epsilon_{ijk} X_k^L, \qquad [X_i^R, X_j^R] = \epsilon_{ijk} X_k^R, \qquad [X_i^L, X_j^R] = 0.$$

Moreover $J_k = X_k^L + X_k^R$ and $K_k = i(X_k^L - X_k^R)$.

24.4 Classification of finite-dimensional representations

For $m \geq 0$ let $V_m = \mathrm{Sym}^m(\mathbb{C}^2)$, and let $S^m(A)$ be the restriction of $A^{\otimes m}$ to V_m. The entries of $S^m(A)$ are polynomials in the entries of A, so dS^m is complex linear on $\mathfrak{sl}(2,\mathbb{C})$. From the Lie algebra background, dS^m is the irreducible module of highest weight m. The element $H = \mathrm{diag}(1,-1)$ acts with eigenvalues $m, m-2, \dots, -m$. On $\mathrm{SU}(2)$, V_m is the spin-$m/2$ representation.

Definition 24.8. For $j_L, j_R \in \{0, \frac{1}{2}, 1, \frac{3}{2}, \dots\}$, the representation (j_L, j_R) of $\mathrm{SL}(2,\mathbb{C})$ is

$$D^{(j_L, j_R)}(A) = S^{2j_L}(A) \otimes S^{2j_R}(\bar{A}) \quad \text{on} \quad V_{2j_L} \otimes V_{2j_R}.$$

Since $A \mapsto \bar{A}$ is a continuous automorphism of $\mathrm{SL}(2,\mathbb{C})$, this is a continuous representation. Its dimension is $(2j_L + 1)(2j_R + 1)$.

A representation Π of $\mathrm{SL}(2,\mathbb{C})$ *descends* to $\mathrm{SO}^+(1,3)$ if there is a continuous representation Π' of $\mathrm{SO}^+(1,3)$ with

$$\Pi = \Pi' \circ \varphi.$$

This is the descent of Lecture 4, now through the covering φ.

Theorem 24.9. *(a) Each (j_L, j_R) is irreducible, and distinct pairs give inequivalent representations. Every irreducible finite-dimensional continuous complex representation of $\mathrm{SL}(2,\mathbb{C})$ is equivalent to some (j_L, j_R).*

(b) $D^{(j_L, j_R)}(-I) = (-1)^{2j_L + 2j_R}$. Hence (j_L, j_R) descends to $\mathrm{SO}^+(1,3)$ if and only if $2j_L + 2j_R$ is even. Up to equivalence, these descended representations are all the irreducible finite-dimensional continuous complex representations of $\mathrm{SO}^+(1,3)$.

(c) The restriction of (j_L, j_R) to $\mathrm{SU}(2)$ is equivalent to $\bigoplus_j V_{2j}$, where $j = |j_L - j_R|, |j_L - j_R| + 1, \dots, j_L + j_R$.

Proof. (a) Write $D = D^{(j_L, j_R)}$, $m = 2j_L$ and $n = 2j_R$. Since $\overline{\exp tY} = \exp t\bar{Y}$, differentiation gives

$$dD(Y) = dS^m(Y) \otimes 1 + 1 \otimes dS^n(\bar{Y}), \qquad Y \in \mathfrak{sl}(2,\mathbb{C})_{\mathbb{R}}.$$

So the complex-linear extension is

$$dD_{\mathbb{C}}(Y \otimes z) = dS^m(zY) \otimes 1 + 1 \otimes dS^n(z\bar{Y}) = (dS^m \boxtimes dS^n)\big(\iota(Y \otimes z)\big).$$

Thus, through ι, $dD_{\mathbb{C}}$ is the module $V_m \boxtimes V_n$ of $\mathfrak{sl}(2,\mathbb{C}) \oplus \mathfrak{sl}(2,\mathbb{C})$. Here $(Y_1, Y_2) \in \mathfrak{sl}(2,\mathbb{C}) \oplus \mathfrak{sl}(2,\mathbb{C})$ acts on $V_m \boxtimes V_n = V_m \otimes V_n$ by $dS^m(Y_1) \otimes 1 + 1 \otimes dS^n(Y_2)$. We import from the prerequisite course Lie Algebras (Lectures 13 and 26): these modules are irreducible and pairwise non-isomorphic, since their highest weights (m, n) differ. Every finite-dimensional irreducible module of $\mathfrak{sl}(2,\mathbb{C}) \oplus \mathfrak{sl}(2,\mathbb{C})$ is one of them.

Now we pass from algebra to group. Every continuous finite-dimensional representation Π is smooth (Lecture 3), with $\Pi(\exp Y) = \exp d\Pi(Y)$ (Lecture 2). By Step 2 of the proof of Theorem 24.5, $\mathrm{SL}(2,\mathbb{C})$ is connected, so it is generated by $\exp \mathfrak{sl}(2,\mathbb{C})$ (Lecture 2). Hence a subspace is Π-invariant if and only if it is $d\Pi$-invariant. A linear map intertwines two group representations if and only if it intertwines their differentials. By the extension facts in § 24.3, both statements pass to $d\Pi_{\mathbb{C}}$. So D is irreducible, and distinct pairs are inequivalent.

Finally, let Π be an irreducible finite-dimensional continuous complex representation. Then $d\Pi_{\mathbb{C}} \circ \iota^{-1} \cong V_m \boxtimes V_n$ for some (m, n). This is the complexified differential of $D^{(m/2, n/2)}$, so $\Pi \cong D^{(m/2, n/2)}$.

(b) We have $S^m(-I) = (-1)^m$ and $\overline{-I} = -I$, so $D(-I) = (-1)^{m+n}$. Covering maps are open. So by the double-cover theorem, Theorem 24.5, φ identifies $\mathrm{SO}^+(1,3)$ with $\mathrm{SL}(2,\mathbb{C})/\{\pm I\}$ as topological groups. By Lecture 4, continuous representations of $\mathrm{SO}^+(1,3)$ correspond to continuous representations of $\mathrm{SL}(2,\mathbb{C})$ that are trivial on $-I$. The correspondence preserves irreducibility and equivalence. Now combine this with (a).

(c) Let $\varepsilon = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$. For $U \in \mathrm{SU}(2)$ one has $\bar{U} = \varepsilon U \varepsilon^{-1}$ (Exercise 24.3). So $S^n(\bar{U}) = S^n(\varepsilon) S^n(U) S^n(\varepsilon)^{-1}$, and the restriction is equivalent to $V_m \otimes V_n$. The Clebsch–Gordan rule from the prerequisite course Lie Algebras (Lectures 6 and 27) gives the decomposition. ∎

Every finite-dimensional continuous complex representation of $\mathrm{SL}(2,\mathbb{C})$ is a direct sum of representations (j_L, j_R). This follows from Weyl's complete reducibility theorem for the semisimple algebra $\mathfrak{sl}(2,\mathbb{C}) \oplus \mathfrak{sl}(2,\mathbb{C})$, transferred as in the proof.

The representations $(j, 0)$ are the holomorphic extensions from $\mathrm{SU}(2)$ given by Lecture 14, since the entries of $S^{2j}(A)$ are polynomials in the entries of A.

Only $(0, 0)$ admits an invariant inner product. The reason is that a boost generator has the nonzero real eigenvalue $j_L + j_R$ on every $(j_L, j_R) \neq (0, 0)$. Exercise 24.6 gives the proof. So every nontrivial continuous unitary representation of this group is infinite-dimensional. Such representations exist. They are induced from the upper triangular subgroup, as in Lecture 22, and their models use the sphere of light rays. We do not study them; see Knapp, *Representation Theory of Semisimple Groups*, Chapter II, Section 4.

In physics, let a field ψ take values in a representation space V with matrices $D(A)$. Its transformation is

$$\psi'(x) = D(A)\psi(\varphi(A)^{-1}x).$$

The label (j_L, j_R) classifies the components of ψ. It does not describe the unitary Hilbert space of particle states, which is the subject of Lecture 25.

24.5 Weyl spinors and four-vectors

Example 24.10 (Weyl spinors)**.** The representation $(\frac{1}{2}, 0)$ is $\mathbb{C}^2$ with $A \mapsto A$. The representation $(0, \frac{1}{2})$ is $\mathbb{C}^2$ with $A \mapsto \bar{A}$. Physicists call their vectors left-handed and right-handed Weyl spinors. The choice of hands is a convention. By the classification theorem, Theorem 24.9, the two representations are inequivalent. A direct check uses traces. For $A = \operatorname{diag}(2i, -\frac{i}{2}) \in \mathrm{SL}(2, \mathbb{C})$,

$$\operatorname{tr} A = \tfrac{3i}{2}, \qquad \operatorname{tr} \bar{A} = -\tfrac{3i}{2}.$$

Equivalent representations have equal traces. These traces differ, so the trace check confirms that the two representations are inequivalent. Both representations restrict to V_1 on $\mathrm{SU}(2)$. Both send $-I$ to -1, so neither descends to $\mathrm{SO}^+(1,3)$. The Dirac representation $(\frac{1}{2}, 0) \oplus (0, \frac{1}{2})$ also sends $-I$ to -1, so it does not descend either.

Example 24.11 (Four-vectors)**.** Let $\Psi : \mathbb{C}^2 \otimes \mathbb{C}^2 \to M_2(\mathbb{C})$ be the linear isomorphism given by $\Psi(\xi \otimes \zeta) = \xi\zeta^T$. The map Ψ sends $e_i \otimes e_j$ to the matrix unit E_{ij}. For $A \in \mathrm{SL}(2, \mathbb{C})$,

$$\Psi(A\xi \otimes \bar{A}\zeta) = A\xi(\bar{A}\zeta)^T = A\xi\zeta^T\bar{A}^T = A\,\Psi(\xi \otimes \zeta)\,A^*.$$

So $D^{(\frac{1}{2},\frac{1}{2})}(A)$ becomes $Z \mapsto AZA^*$ on $M_2(\mathbb{C}) = \mathcal{H} \oplus i\mathcal{H}$. This action preserves $\mathcal{H}$ and acts there by $\varphi(A)$. Hence $(\frac{1}{2}, \frac{1}{2})$ is equivalent to $\mathbb{C}^4 = \mathbb{R}^4 \otimes_{\mathbb{R}} \mathbb{C}$ with $\varphi(A)$ extended complex-linearly. Since $2 \cdot \frac{1}{2} + 2 \cdot \frac{1}{2} = 2$ is even, the representation $(\frac{1}{2}, \frac{1}{2})$ descends by the descent criterion of Theorem 24.9. The descended representation is the defining representation of $\mathrm{SO}^+(1,3)$ on $\mathbb{C}^4$.

Two checks are consistent with this identification. First, the restriction rule of Theorem 24.9(c) predicts $V_0 \oplus V_2$ on $\mathrm{SU}(2)$. Indeed $U \in \mathrm{SU}(2)$ acts by $Z \mapsto UZU^{-1}$. It fixes I, which spans the time direction. It preserves the traceless matrices, which are the spatial vectors and carry spin 1. Second, compare boost eigenvalues. In $(\frac{1}{2}, \frac{1}{2})$, the element $iX_3 = \frac{1}{2}\sigma_3$ acts by $\frac{1}{2}(H \otimes 1 + 1 \otimes H)$. Its eigenvalues are $\frac{1}{2}(q + q')$ with $q, q' \in \{\pm 1\}$:

$$1, \quad 0, \quad 0, \quad -1.$$

On $\mathbb{R}^4$, the matrix $d\varphi(iX_3) = K_3$ has eigenvectors $e_0 + e_3$, e_1, e_2, $e_0 - e_3$ with eigenvalues $1, 0, 0, -1$. The lists agree.

The table lists small cases. Exercise 24.4 identifies $(1,0) \oplus (0,1)$ with $\Lambda^2\mathbb{C}^4$, the complexified antisymmetric tensors.

(j_L, j_R)	dimension	on $\mathrm{SU}(2)$	descends	name
$(0,0)$	1	V_0	yes	scalar
$(\frac{1}{2},0)$, $(0,\frac{1}{2})$	2	V_1	no	Weyl spinors
$(1,0)$, $(0,1)$	3	V_2	yes	summands of $\Lambda^2\mathbb{C}^4$
$(\frac{1}{2},\frac{1}{2})$	4	$V_0 \oplus V_2$	yes	four-vector
$(1,\frac{1}{2})$	6	$V_1 \oplus V_3$	no	–

In physics language, the two summands of $\Lambda^2\mathbb{C}^4$ are spanned by the complex combinations $\vec{E} + i\vec{B}$ and $\vec{E} - i\vec{B}$ of the electric and magnetic fields. Which one is called $(1,0)$ depends on the convention for hands.

24.6 Self-checks

(1) Compute $\varphi(\operatorname{diag}(2, \tfrac{1}{2}))$.
(2) Find all $A \in \mathrm{SL}(2,\mathbb{C})$ with $\varphi(A) = R_3(\pi)$.
(3) Does $(\tfrac{3}{2}, \tfrac{1}{2})$ descend to $\mathrm{SO}^+(1,3)$? Give its dimension and its restriction to $\mathrm{SU}(2)$.

Answer. (1) By the diagonal formula of Example 24.4 with $a = 2$, we get $x_0 \pm x_3 \mapsto 4^{\pm 1}(x_0 \pm x_3)$. Since $a/\bar{a} = 1$, the coordinates x_1 and x_2 are fixed. This is $B_3(t)$ with $e^t = 4$, so $\cosh t = \frac{17}{8}$ and $\sinh t = \frac{15}{8}$. Thus

$$\varphi\big(\operatorname{diag}(2, \tfrac{1}{2})\big) = \begin{pmatrix} \frac{17}{8} & 0 & 0 & \frac{15}{8} \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ \frac{15}{8} & 0 & 0 & \frac{17}{8} \end{pmatrix}.$$

Check: $17^2 - 15^2 = 64$.

(2) The element $\exp(\pi X_3) = \operatorname{diag}(-i, i)$ maps to $R_3(\pi)$. The fibre is a coset of $\{\pm I\}$, so the answer is $A = \pm\operatorname{diag}(-i, i)$.

(3) Here $2j_L + 2j_R = 4$ is even, so it descends. Its dimension is $4 \cdot 2 = 8$. The restriction is $V_2 \oplus V_4$, of dimension $3 + 5 = 8$.

24.7 Summary

The action $X \mapsto AXA^*$ on Hermitian matrices preserves $\det X = \langle x, x\rangle$. It gives a two-sheeted covering $\varphi : \mathrm{SL}(2,\mathbb{C}) \to \mathrm{SO}^+(1,3)$ with kernel $\{\pm I\}$. Its restriction to real matrices is the double cover $\mathrm{SL}(2,\mathbb{R}) \to \mathrm{SO}^+(1,2)$. The map ι identifies the complexified Lorentz algebra with $\mathfrak{sl}(2,\mathbb{C}) \oplus \mathfrak{sl}(2,\mathbb{C})$. Up to equivalence, the irreducible finite-dimensional continuous representations of $\mathrm{SL}(2,\mathbb{C})$ are the (j_L, j_R), given by $S^{2j_L}(A) \otimes S^{2j_R}(\bar{A})$. Exactly those with $2j_L + 2j_R$ even descend to $\mathrm{SO}^+(1,3)$. Weyl spinors are $(\tfrac{1}{2}, 0)$ and $(0, \tfrac{1}{2})$, and four-vectors are $(\tfrac{1}{2}, \tfrac{1}{2})$.

24.8 Reading

- A. W. Knapp, *Representation Theory of Semisimple Groups: An Overview Based on Examples*, Princeton University Press, 2001, Chapter II, Section 3, on finite-dimensional representations of $\mathrm{SL}(2,\mathbb{C})$.